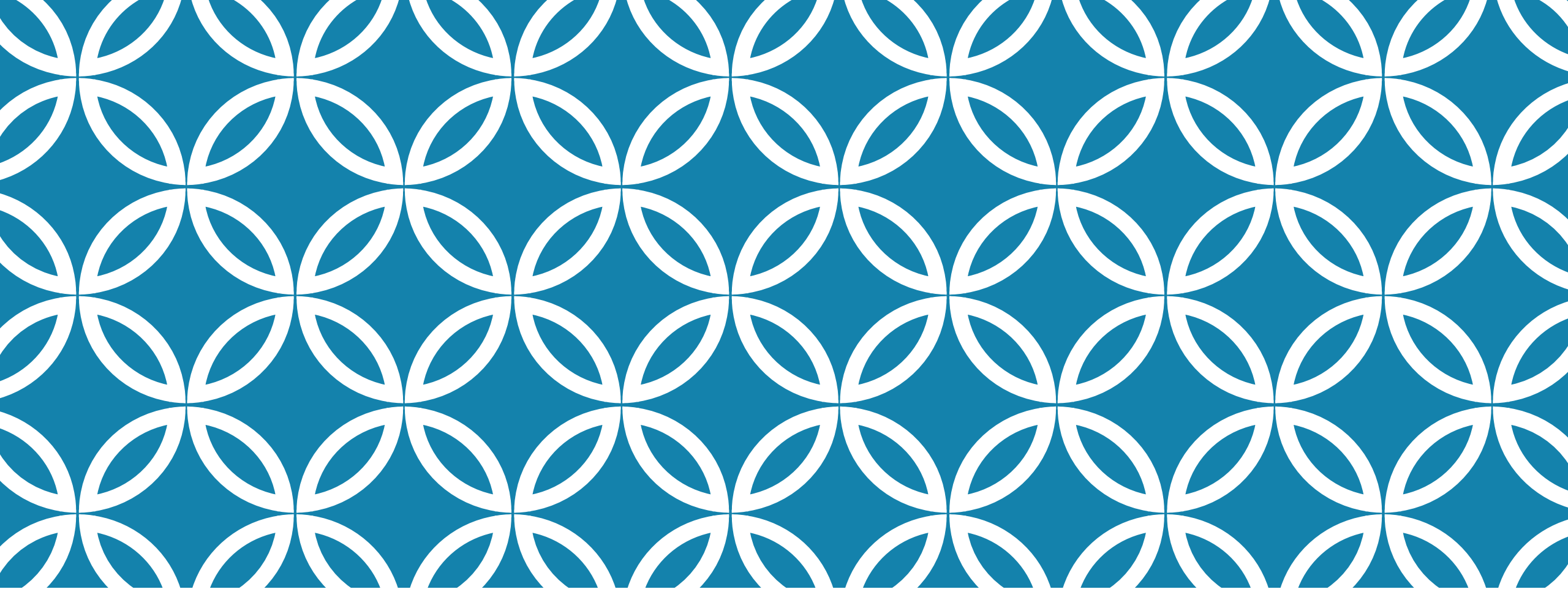




SEABORN

Eng: Asmaa Farhat

SEABORN

- Seaborn is a library for making statistical graphics in Python. It builds on top of [matplotlib](#) and integrates closely with [pandas](#) data structures.
- Seaborn helps you explore and understand your data. Its plotting functions operate on dataframes and arrays containing whole datasets

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Tips dataset

	A	B	C	D	E	F	G	H
1	total_bill	tip	sex	smoker	day	time	size	
2	16.99	1.01	Female	No	Sun	Dinner	2	
3	10.34	1.66	Male	No	Sun	Dinner	3	
4	21.01	3.5	Male	No	Sun	Dinner	3	
5	23.68	3.31	Male	No	Sun	Dinner	2	
6	24.59	3.61	Female	No	Sun	Dinner	4	
7	25.29	4.71	Male	No	Sun	Dinner	4	
8	8.77	2	Male	No	Sun	Dinner	2	
9	26.88	3.12	Male	No	Sun	Dinner	4	
10	15.04	1.96	Male	No	Sun	Dinner	2	
11	14.78	3.23	Male	No	Sun	Dinner	2	
12	10.27	1.71	Male	No	Sun	Dinner	2	
13	35.26	5	Female	No	Sun	Dinner	4	
14	15.42	1.57	Male	No	Sun	Dinner	2	

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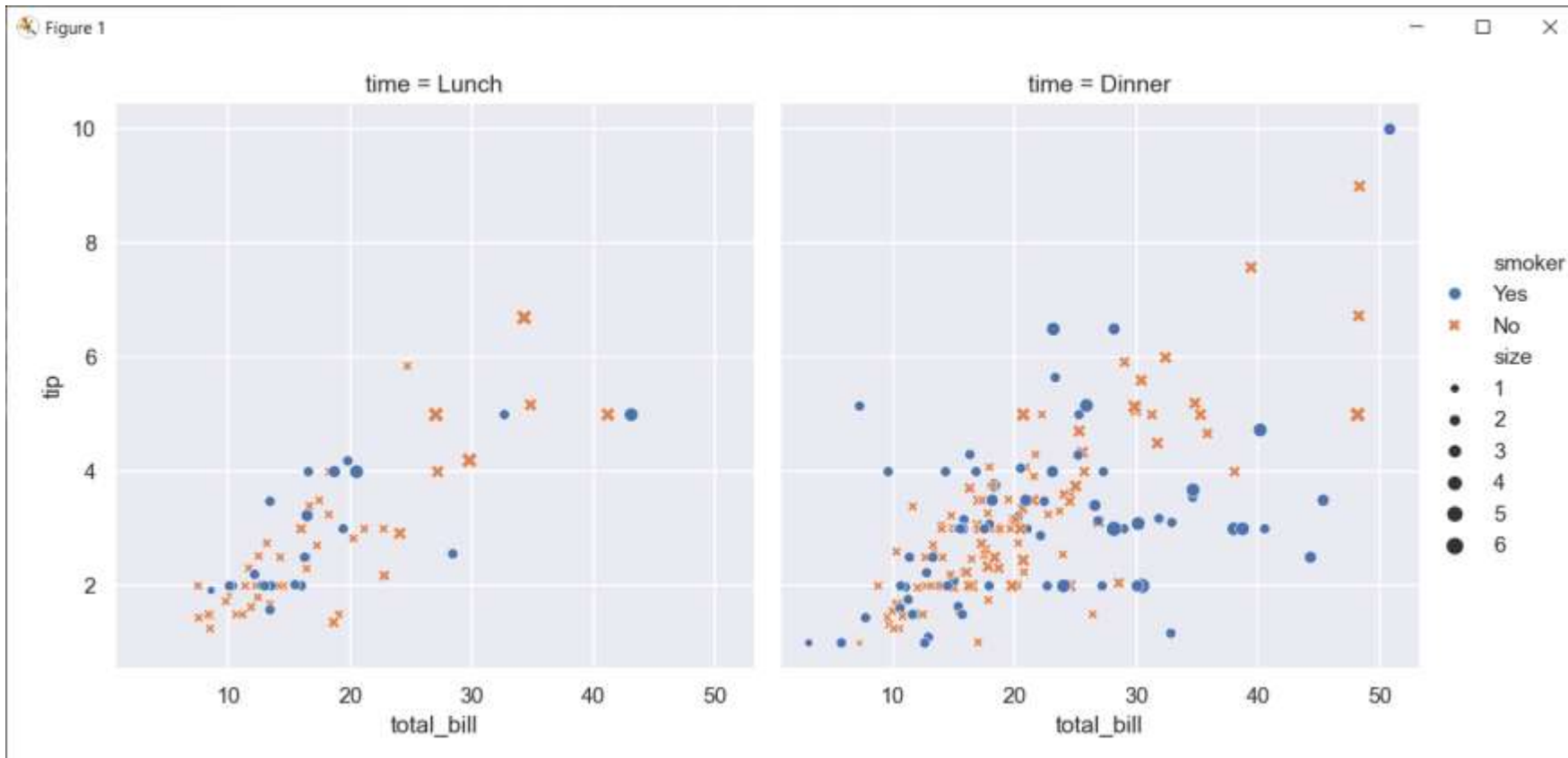
```
# Import seaborn
import seaborn as sns

# Apply the default theme
sns.set_theme()

# Load an example dataset
tips = sns.load_dataset("tips")

# Create a visualization
sns.relplot(
    data=tips,
    x="total_bill", y="tip", col="time",
    hue="smoker", style="smoker", size="size",
)
```

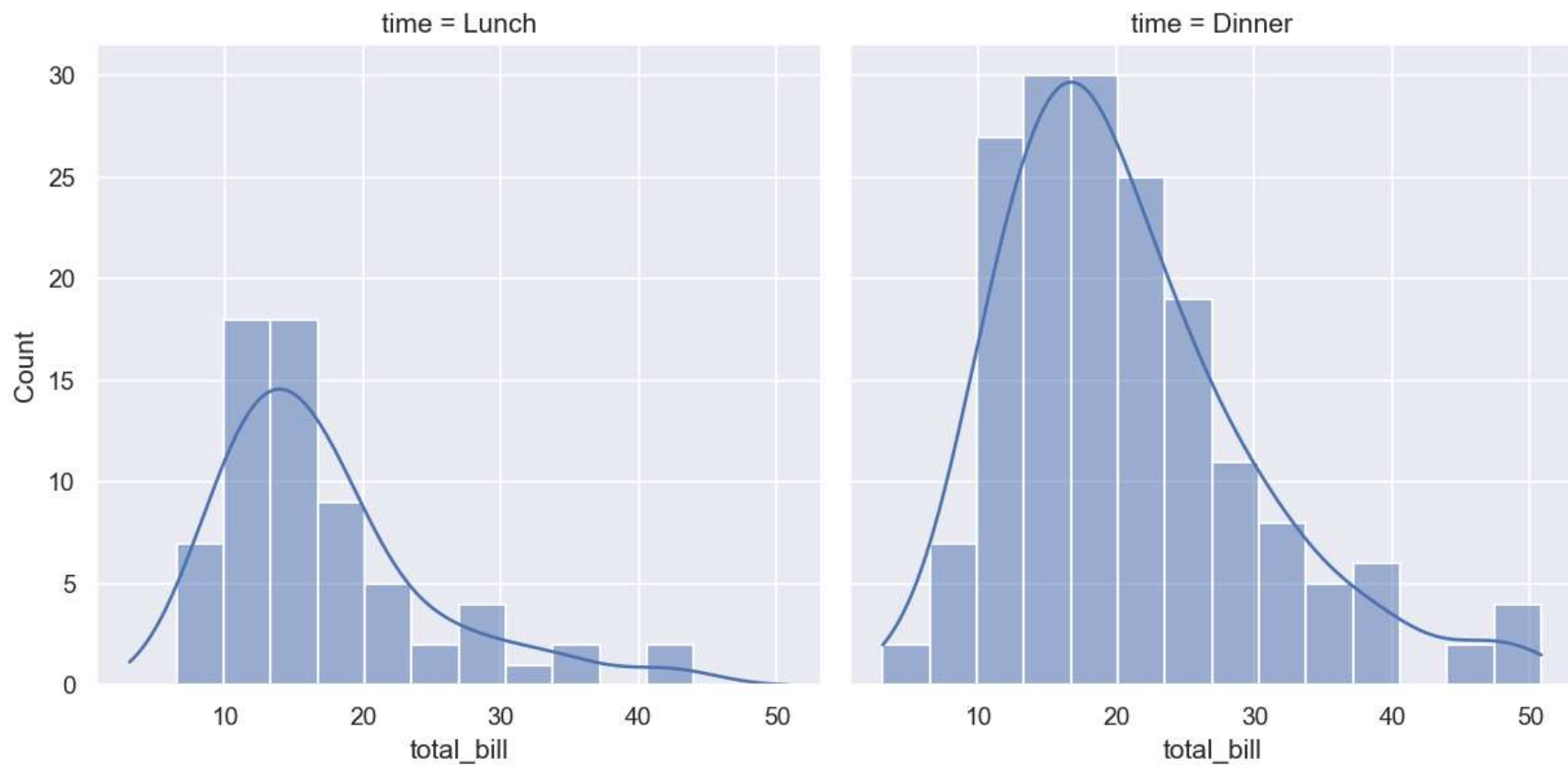
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Distributional representations

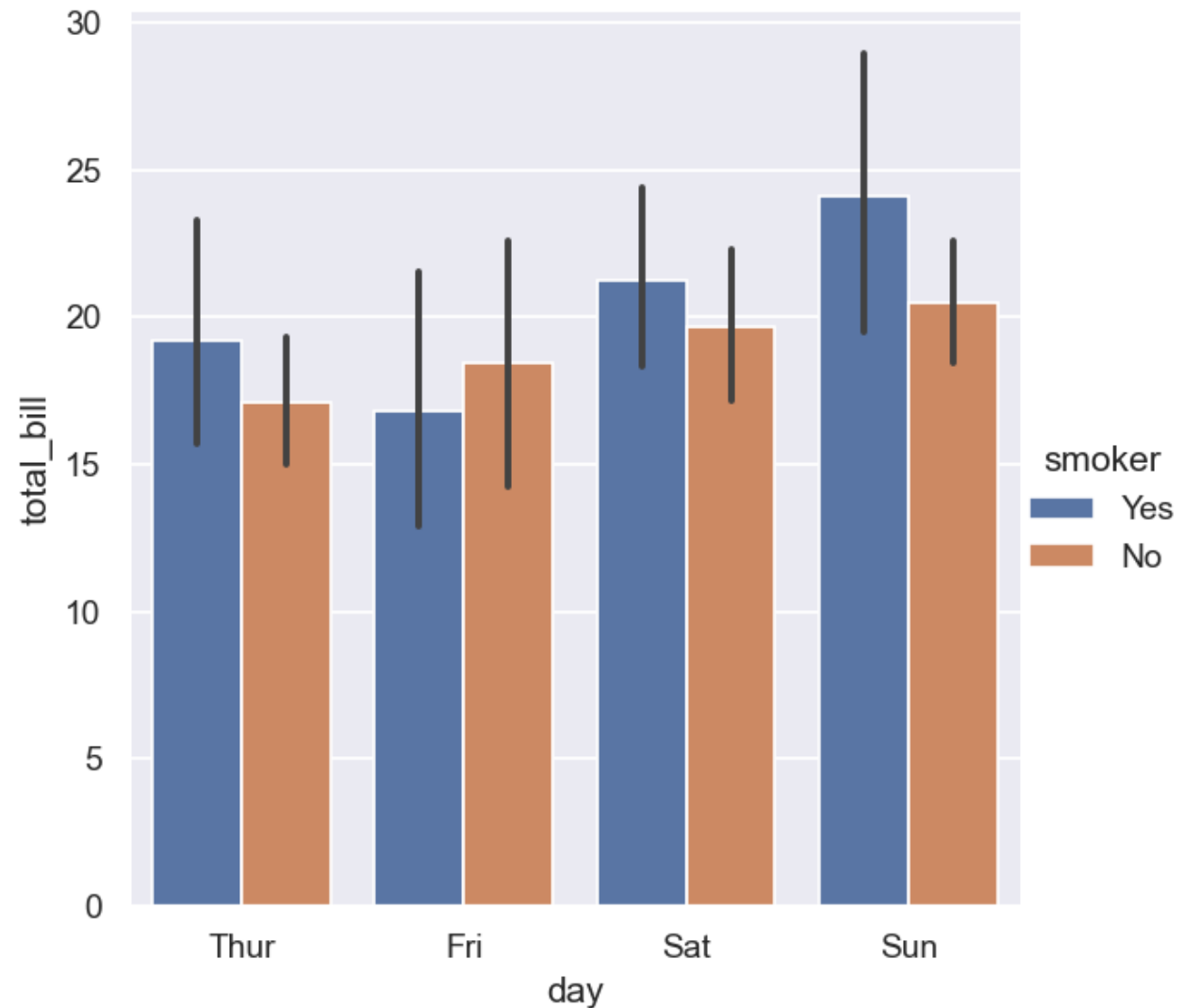
```
sns.displot(data=tips, x="total_bill", col="time", kde=True)
```



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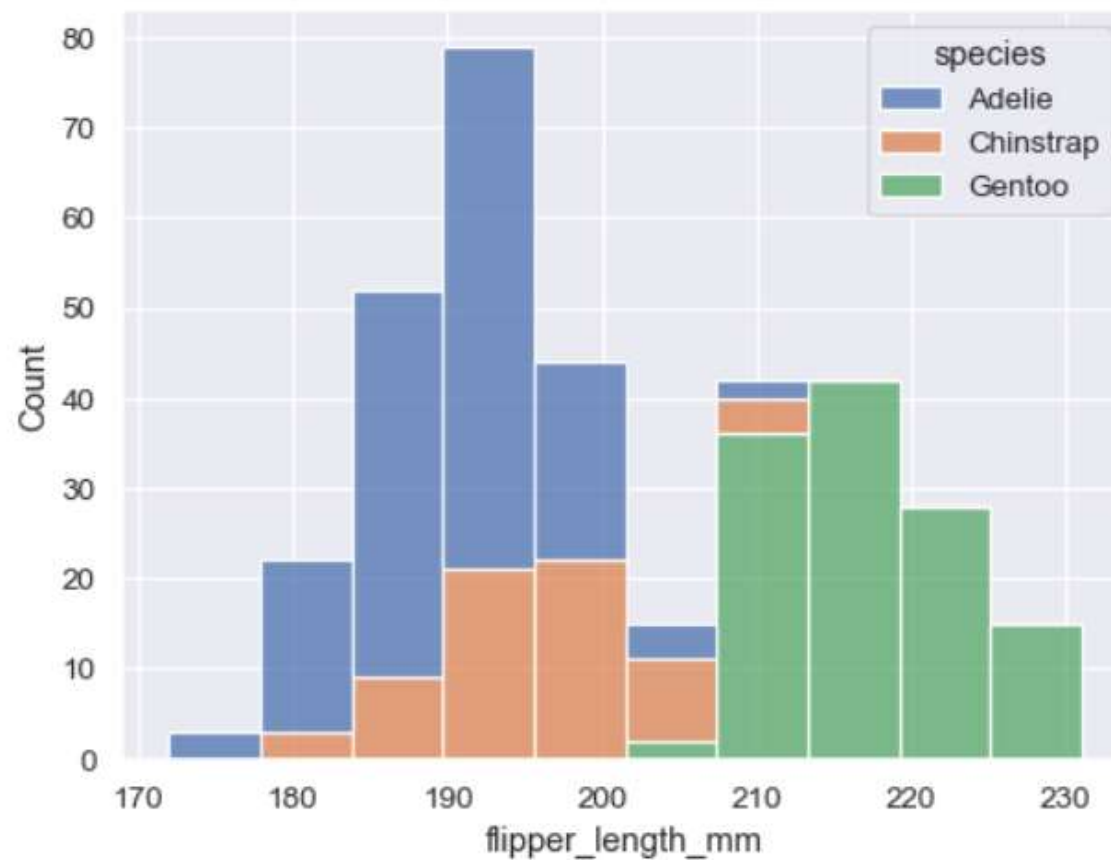
Distributional representations

```
sns.catplot(data=tips, kind="bar", x="day", y="total_bill", hue="smoker")
```



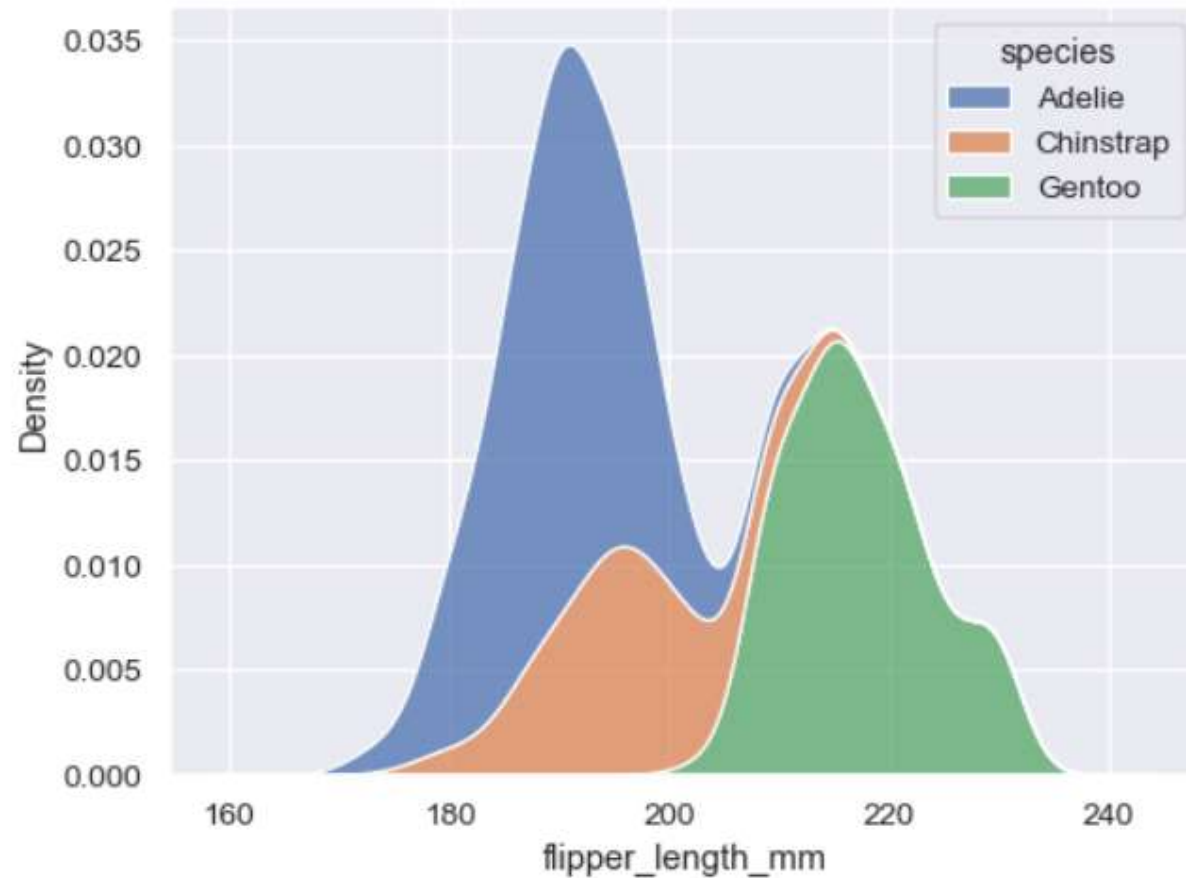
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```
penguins = sns.load_dataset("penguins")  
sns.histplot(data=penguins, x="flipper_length_mm", hue="species", multiple="stack")
```



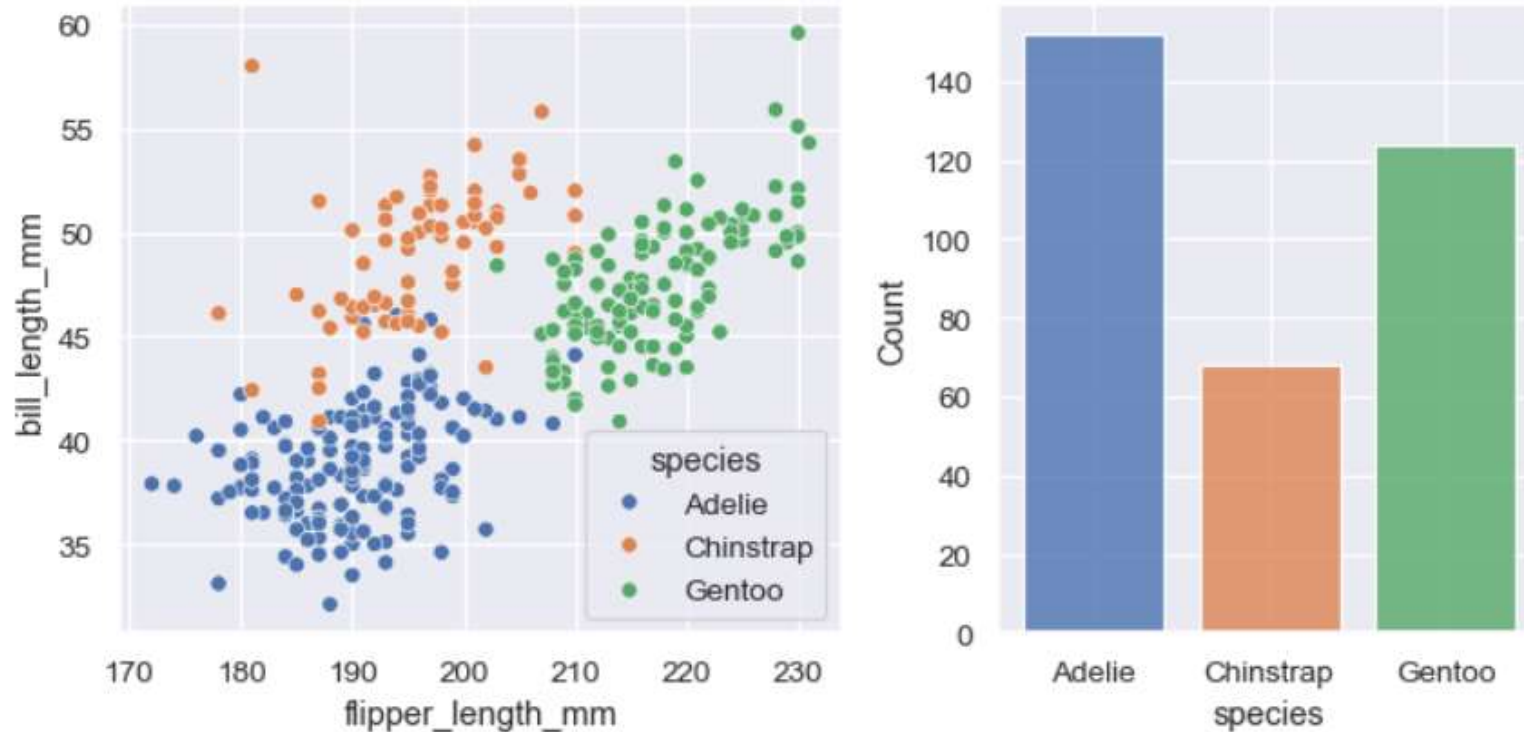
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```
sns.kdeplot(data=penguins, x="flipper_length_mm", hue="species", multiple="stack")
```



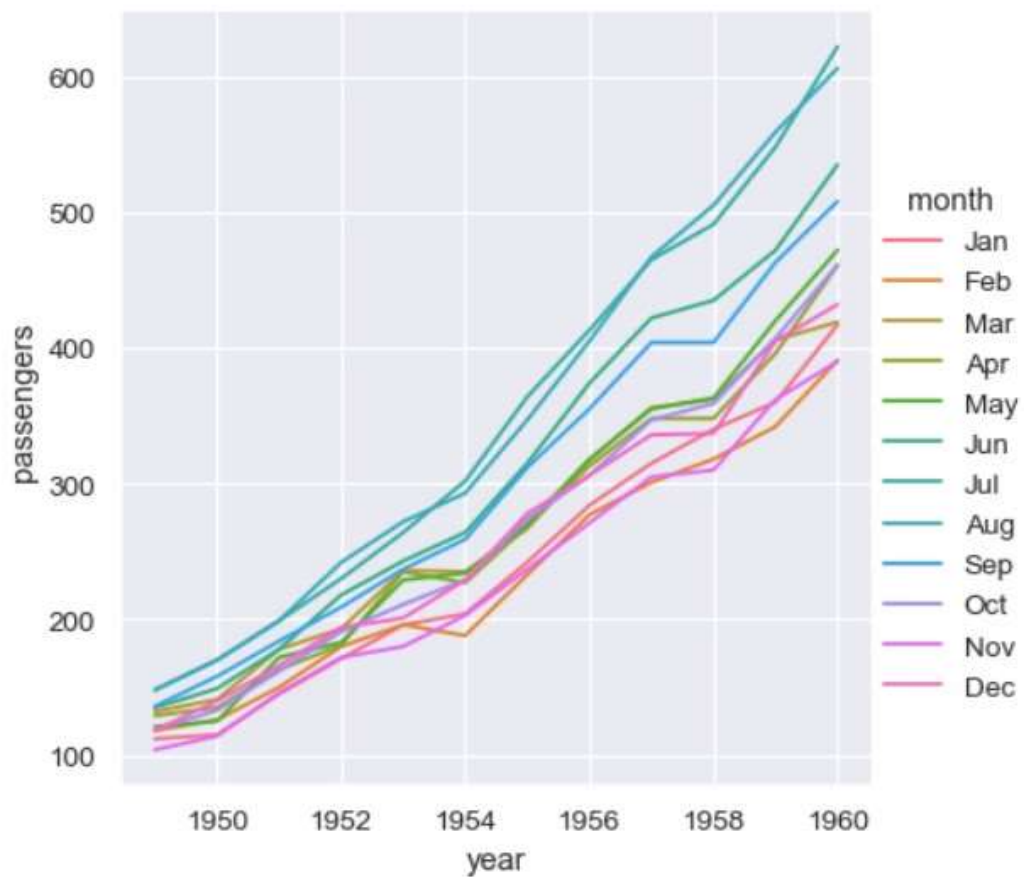
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```
f, axs = plt.subplots(1, 2, figsize=(8, 4), gridspec_kw=dict(width_ratios=[4, 3]))
sns.scatterplot(data=penguins, x="flipper_length_mm", y="bill_length_mm", hue="species", ax=axs[0])
sns.histplot(data=penguins, x="species", hue="species", shrink=.8, alpha=.8, legend=False, ax=axs[1])
f.tight_layout()
```



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```
sns.relplot(data=flights, x="year", y="passengers", hue="month", kind="line")
```



1	year	month	passengers
2	1949	January	112
3	1949	February	118
4	1949	March	132
5	1949	April	129
6	1949	May	121
7	1949	June	135
8	1949	July	148
9	1949	August	148

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```
sns.catplot(data=tips, x="day", y="total_bill", hue="sex", kind="swarm")
```

